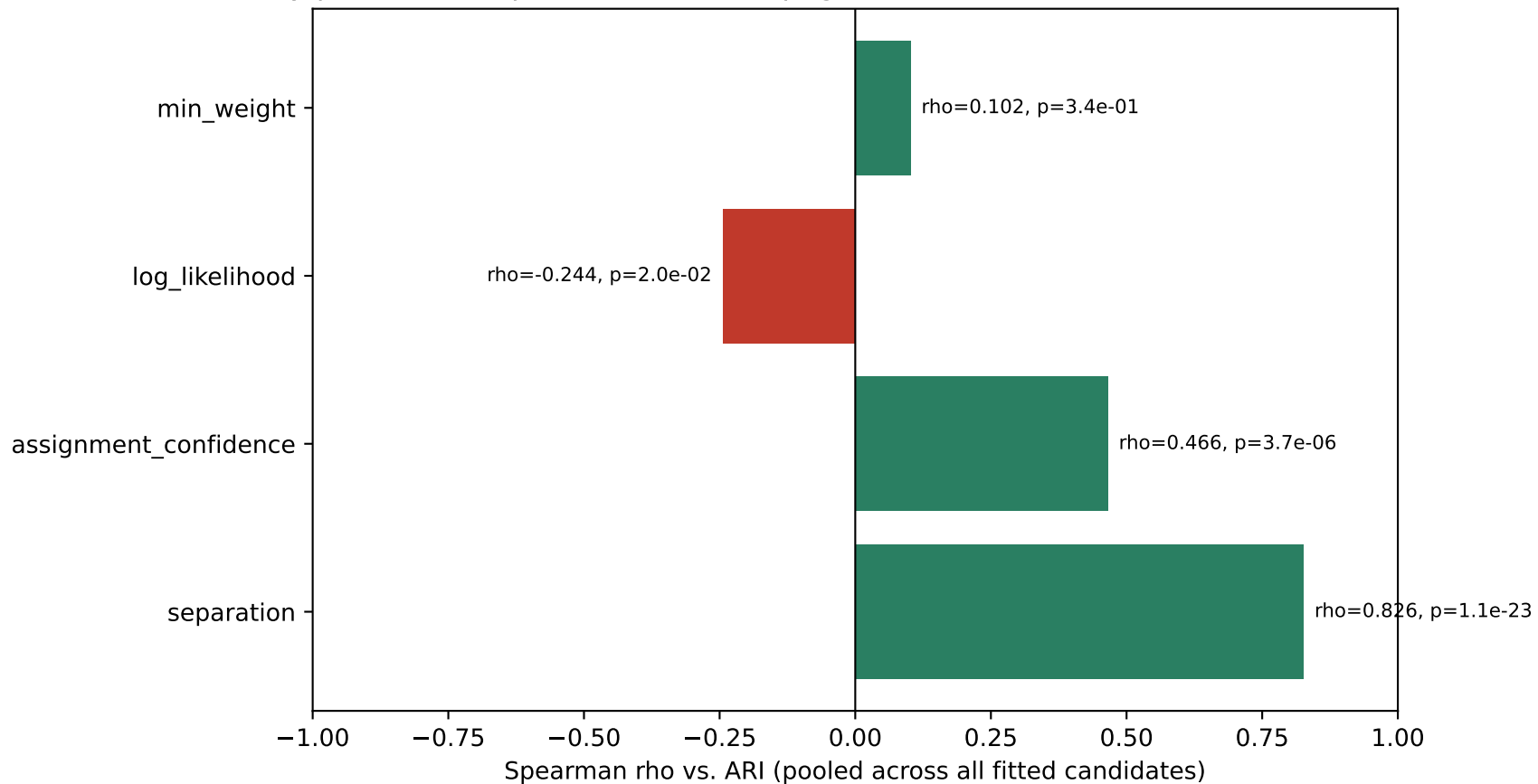
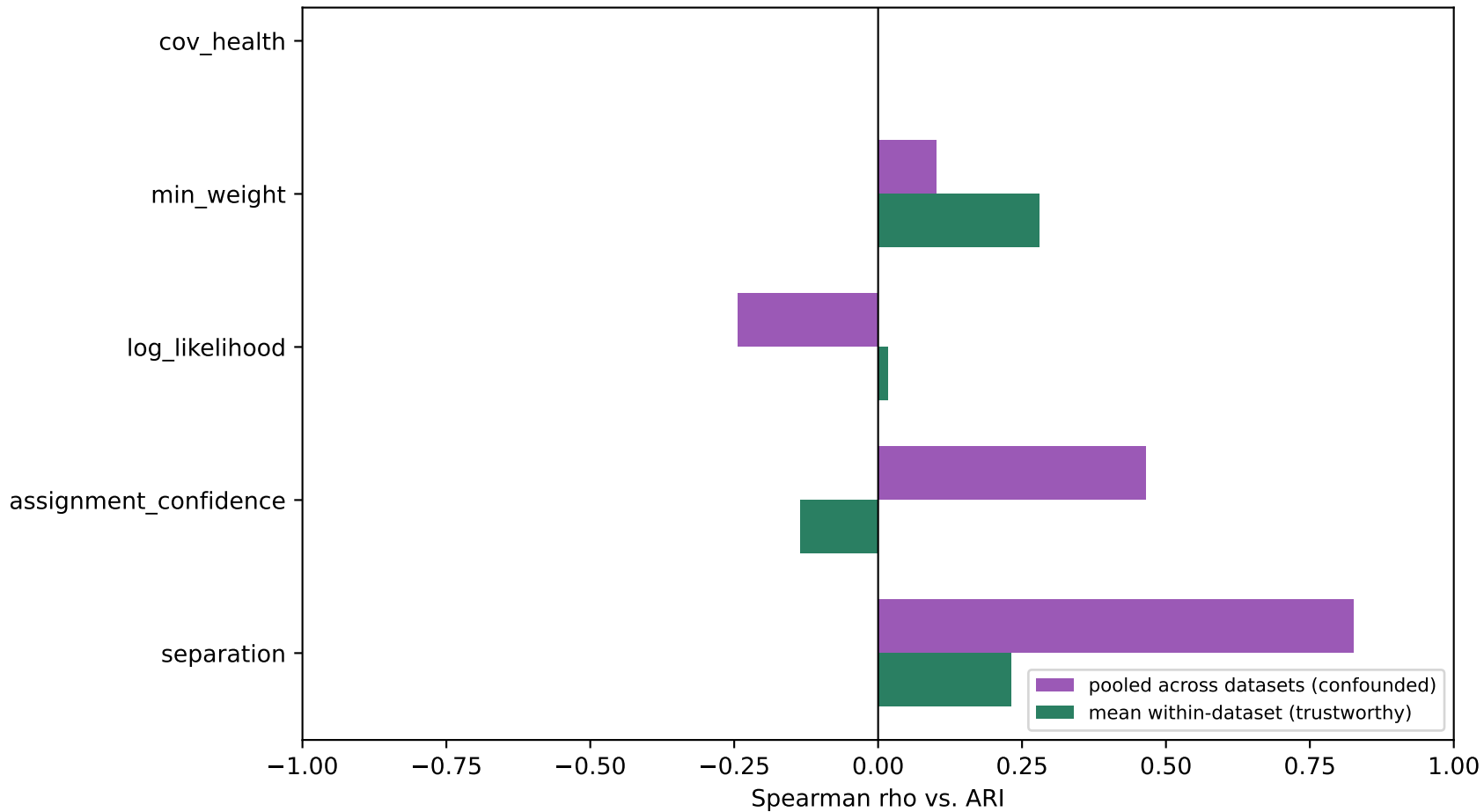


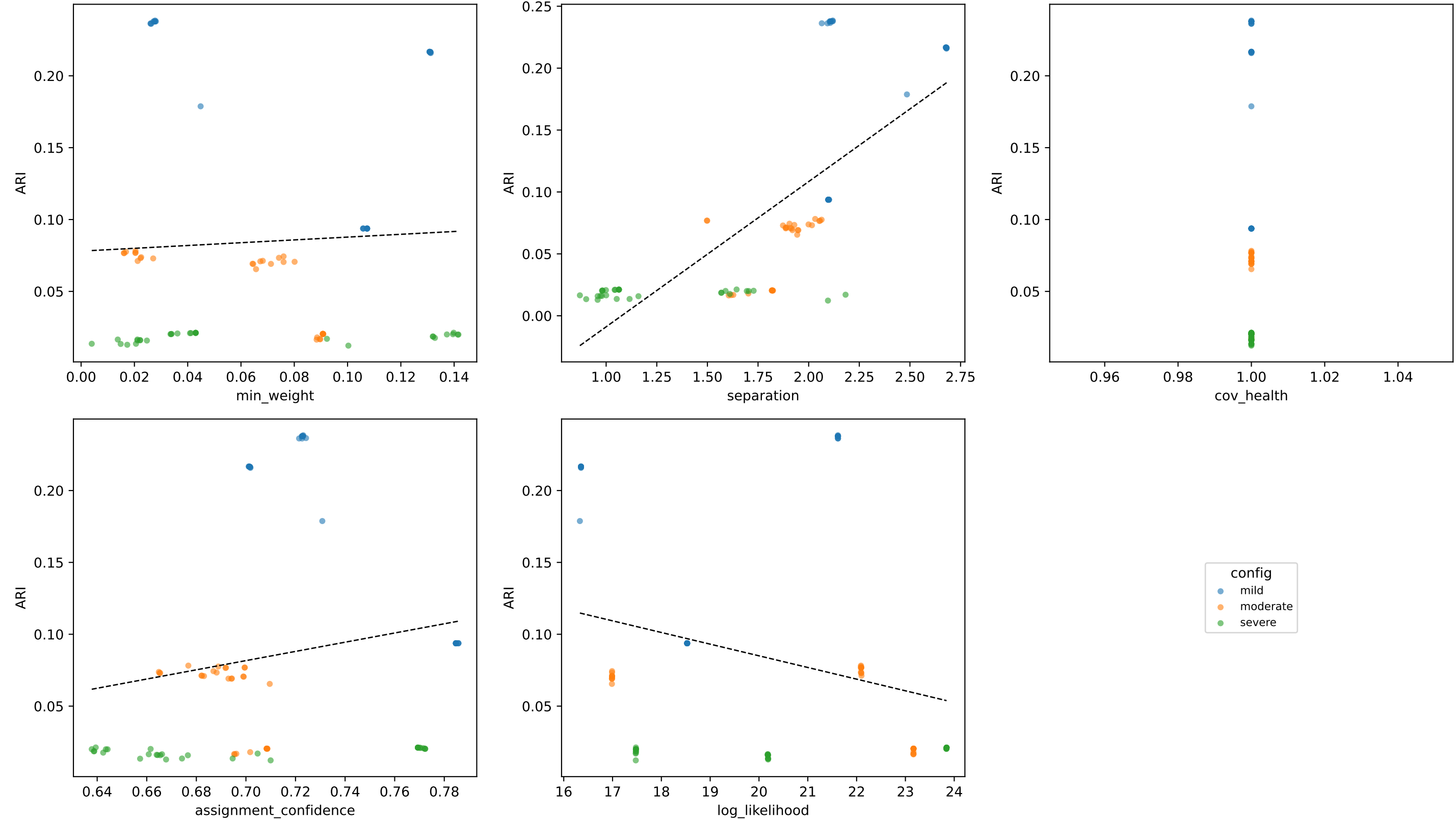
h raw metrics actually predict ARI? (pooled -- see next page: this inflates/deflates metrics via cross-dataset



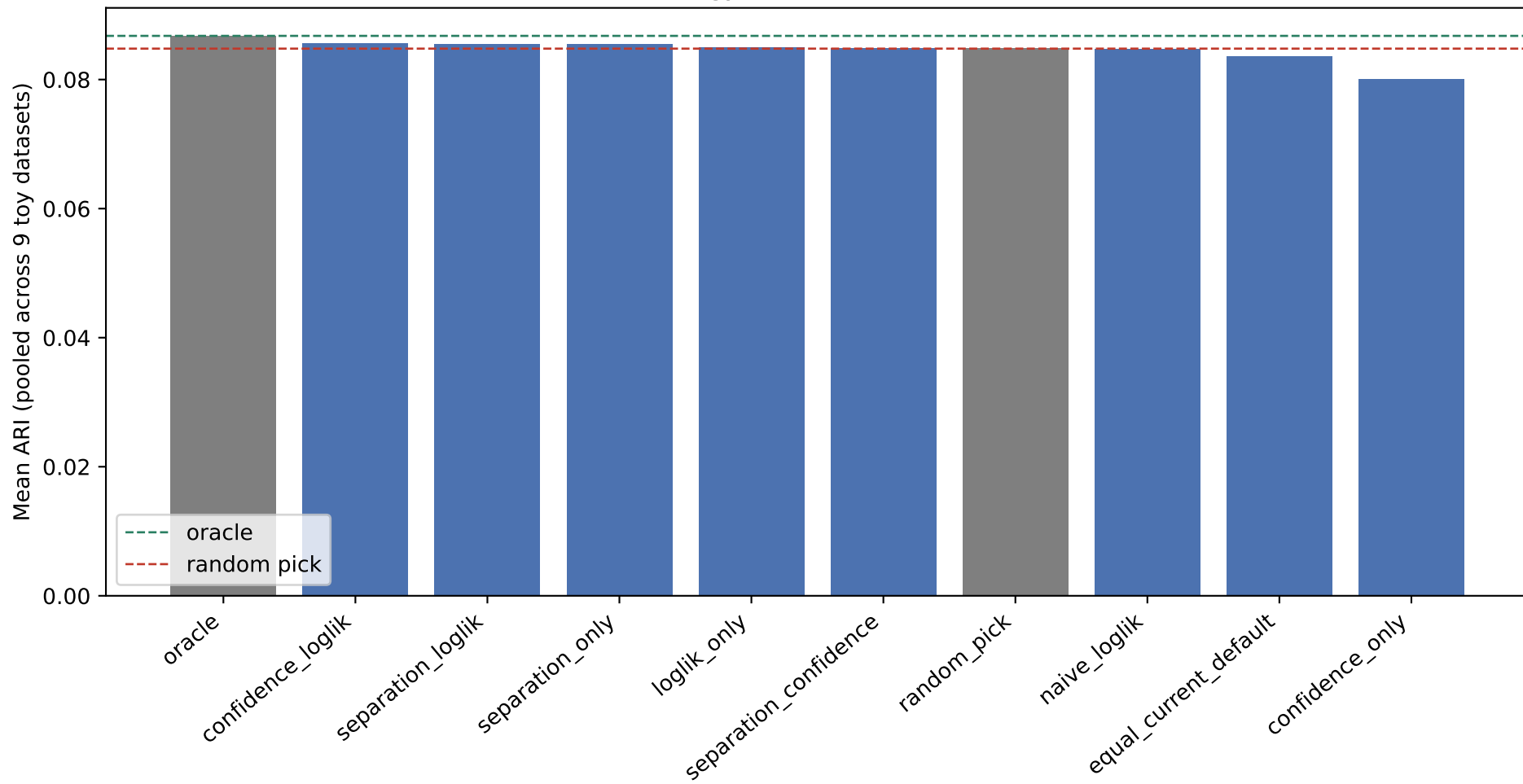
Pooled correlation can overstate (or invert) predictive power --
within-dataset rho is the number to trust



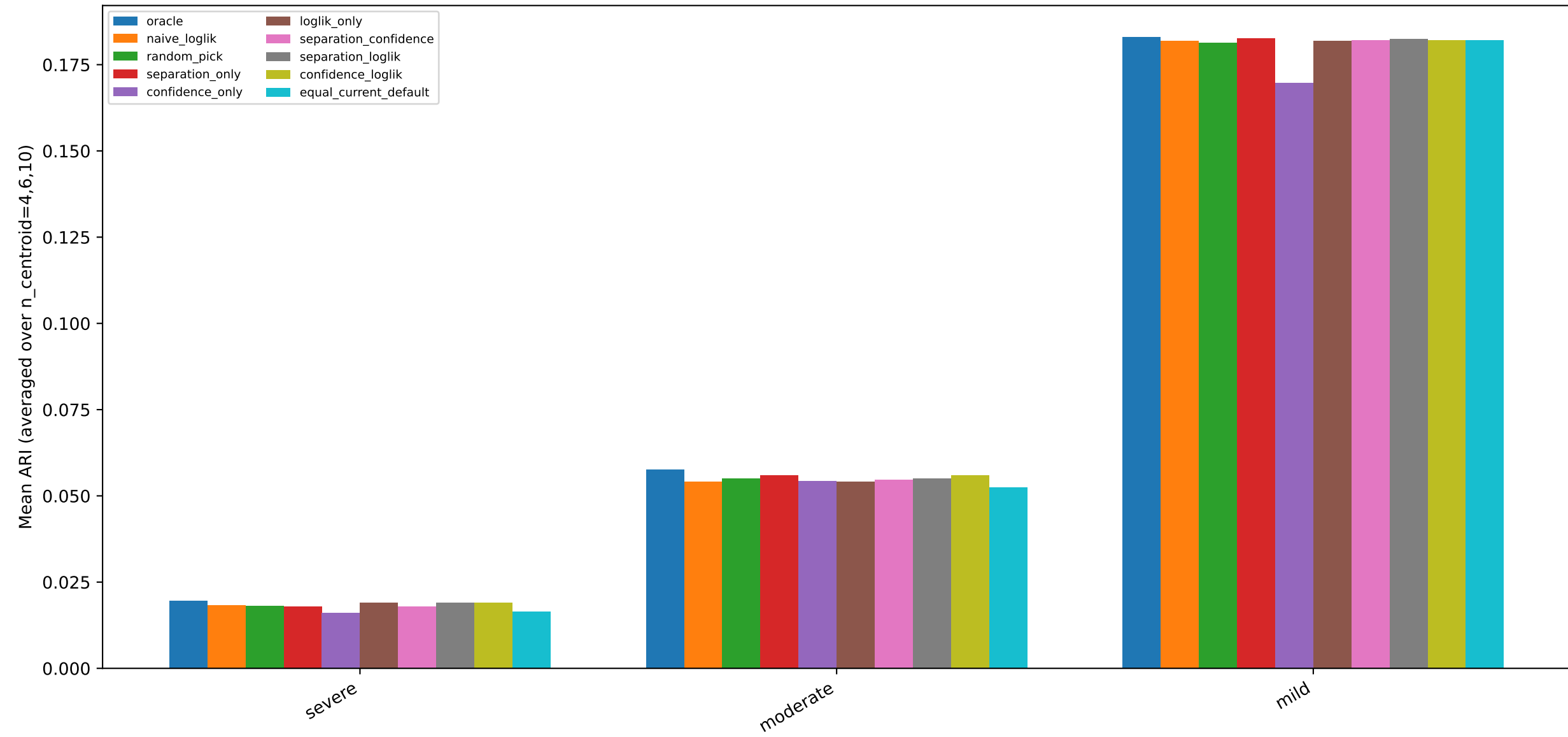
Metric value vs. ARI, per candidate (90 candidates, colored by stress config)



Selection strategy vs. oracle / random baselines



Strategy ARI by stress config — where does the metric choice matter?



Metric selection summary

Metric → ARI correlation ranking (pooled, $|\rho|$ descending):

separation	$\rho=+0.826$	$p=1.1e-23$
assignment_confidence	$\rho=+0.466$	$p=3.7e-06$
log_likelihood	$\rho=-0.244$	$p=2.0e-02$
min_weight	$\rho=+0.102$	$p=3.4e-01$
cov_health	$\rho=+nan$	$p=nan$

Metric → ARI correlation ranking (WITHIN-DATASET mean, $|\rho|$ descending -- trust this one):

min_weight	mean_rho=+0.280	(9/9 datasets)
separation	mean_rho=+0.232	(9/9 datasets)
assignment_confidence	mean_rho=-0.136	(9/9 datasets)
log_likelihood	mean_rho=+0.016	(9/9 datasets)
cov_health	mean_rho=n/a	(0/9 datasets)

Top 3 selection strategies by pooled mean ARI:

oracle	mean ARI=0.0868
confidence_loglik	mean ARI=0.0856
separation_loglik	mean ARI=0.0855

oracle	mean ARI=0.0868	(upper bound)
random_pick	mean ARI=0.0848	(lower bound)